

Milestone 2

1. Project Descriptions & Goals

The goal of this project is to design a self-soothing device for college students. The current leading design is to create a ring that facilitates spinning a pen around the user's finger. The device should fit in a pocket, have at least one moving part, and require no post-printing processing. The device would also ideally mimic the action of spinning a pen in one's hand, fit around the user's pointer finger above the knuckle, must allow a pen to be attached, must allow the pen to follow face or fidget ring, must allow the pen to spin on the axis of pen mount, and allow continuous and smooth movement of the pen.

2. Project Motivation

The design is inspired by the challenges that often arise when spinning a pen between one's fingers. Spinning a pen is a common self-soothing behavior commonly observed among students in college classrooms. Additionally, since nearly every college student carries a pen to class, creating a device that facilitates pen spinning is a practical design choice. The ring's design allows the device to cause little distraction or time spent putting it away during classroom moments that require students' full attention. The spinning aspect of the design also allows the student to hold the pen in writing position when needed, eliminating the need to disengage the device to write, take notes, etc.

3. Plan for Achieving Goals

3.1 Milestone 1 Background Information

To recap, from milestone one, three designs were considered for this ring device. The current, preferred rendition consists of a ring shaped like an infinity symbol, bent into a saddle so that the user's finger slides through the ring's rounded ends of the infinity symbol. The joint point of the infinity symbol does not have an actual cross over like a real infinity symbol or figure 8, but rather two curved lines that approach each other while never intersecting. The ring also has a T-shaped track following the face to allow the pen mount to clip onto, so the pen can be slid along the face. The pen mount allows the pen to spin around the axis of the mount.

3.2 Engineering Design Method of Choice

3.2.1 Iterative Nature Discussion

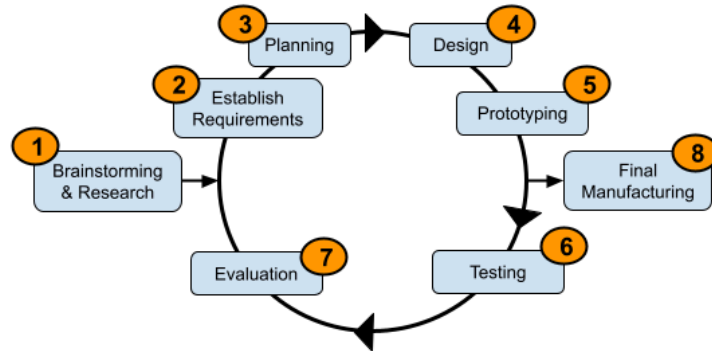


Figure 3.2.1.1: Graphic of the method showing the iterative nature

The figure above outlines the iterative methodology that will guide the development of this device throughout the project. The process commences with box 1, where initial brainstorming and research are conducted, a phase documented as milestone 1. Following this, box 2 marks the stage at which requirements are established. Box 4 represents the phase in which designs are formulated, utilizing both sketch paper and advanced software tools. Subsequently, box 5 indicates the initiation of the construction of initial physical prototypes. Testing and evaluation are conducted after each prototype is developed, and it is important to note that the iterative process spanning from box 2 to box 7 can be repeated as necessary before advancing to box 8.

3.2.2 Cost and Environmental Impact On Iteration

Cost and environmental impact are critical factors that must inform design decisions and modifications throughout the iterative process. Each change necessitates additional iterations, resulting in incremental costs and time commitments. Therefore, the objective of each iteration is not solely to enhance performance but to optimize the balance between improvement and associated impacts. For instance, reinforcing a component may not justify the extra expenditure of 3D filament or the time invested, particularly for compact devices such as the one discussed herein. Furthermore, evaluating the environmental footprint in the early stages of the iteration process is essential to ensure that the final design is both socially responsible and compliant with evolving regulations. By conducting thorough analyses of costs and environmental impacts at the outset, one can limit potentially wasteful expenditures of time and resources in subsequent iterations.

4. Available Resources & Constraints

Table 4.1: Available Resources & Constraints

Available Resources
3D printers at Makerspace & ME 250 lab
Available Fillaments: PLA, ABS, PETG, PCTG, nylon, glass-filled nylon, PA6-CF (carbon fiber nylon), TPU/TPE (various durometers)
SolidWorks software
TA ME 347 lab support hours
Personal Design Notebook
Constraints
Must fit flush around the user's finger.
Must be able to support the weight of a standard pen.
The pen mount must be able to wrap around the diameter of a standard pen.

5. Project Needs

Table 5.1: Needed Software & Hardware

Needed Software
Solidworks
Google Drive (for documentation, drawings, etc.)
Needed Hardware
Design notebook
Standard pen (for testing)
FDM 3D Printer Access
Ring sizing kit (To measure the designer's ring size, ensure the device prints at the correct size.)
Personal computer & access to ME 347 lab computers

6. Project Timeline & Expected Final Deliverables

Table 6.1: Project Timeline

Week	Tasks to Complete
Week 5 (2/9-2/13)	• Milestone 2 • CAD model of device
Week 6 (2/16-2/20)	• Submit prototype 1 for print • Get feedback & make iterations as needed
Week 7 (2/23-2/27)	• Have the final CAD assembly ready to go

Table 6.2: Expected Final Deliverables

Expected Final Deliverables
Fully functional self-soothing device
Unique design (no preexisting patents or similar devices on market)
Completion of various milestones or steps of the project as outlined by the ME 347 curriculum